Gold Price Prediction using Machine Learning

Phase 1 : Discovery Report

CN240 – Data Science for Signal Processing

Computer Engineering, Thammasat School of Engineering

Covering : Overview, Stakeholders, Framing the Problem, Objectives, Success Criteria, Formulating Hypotheses, Data Assessment, Resource Assessment & Roles, Project Risks and Conclusion

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1. Overview

Over the past decade, gold has evolved beyond its traditional role as a safe-haven asset into a complex investment instrument. The global economy during this period has been shaped by persistent uncertainty — from shifting monetary policies and the COVID-19 pandemic to intense geopolitical tensions such as U.S. tariff announcements and ongoing conflicts in the Middle East. These factors have driven gold prices in Thailand to levels of volatility where investment decisions can no longer rely on intuition alone.

This project applies concepts from CN240 – Data Science for Signal Processing to address the challenges of predicting prices in a highly volatile asset market. The year 2026 in particular has been marked by renewed instability from both financial policy changes and geopolitical developments, which have made historical price relationships increasingly difficult to interpret using conventional approaches.

To tackle this, our team built a full Data Science Pipeline — starting from Exploratory Data Analysis (EDA) and Feature Engineering, through to Model Comparison — with two main goals in mind: first, to develop a decision-support tool capable of distinguishing genuine investment signals from market noise; and second, to give every team member hands-on experience across the entire data science workflow.

2. Stakeholders

Since this project was developed for academic purposes — specifically to apply and test data science concepts — the stakeholders can be divided into two main groups.

2.1 Application-Oriented Stakeholders

The primary end-users of this project's findings are retail gold investors. Although the project is academic in nature, the trading signals and feature extraction techniques developed here could serve as a foundation for a basic decision-support tool. This would help individual investors reduce emotional bias and move toward a more data-driven approach when navigating volatile market conditions.

2.2 Academic and Development Stakeholders

The first group consists of the student team members themselves — computer engineering students who aim to apply data science theory to real-world time series data. The goal is to practice and validate end-to-end skills, covering everything from data preprocessing and feature engineering to model development and performance evaluation, across both regression and classification tasks.

The second group is the course instructor, who evaluates the project based on the correctness of the methodology, the reasoning behind model selection — particularly in handling non-stationary data — and the team's ability to interpret results in a way that addresses the research hypotheses.

3. Consider the Past

Studying past projects and related research is a crucial foundation for designing the model architecture for gold price prediction. Learning from previous quantitative tests helps our team avoid using the wrong algorithms and allows us to focus on solutions that have been proven to work in real-world scenarios. The key takeaways can be summarized as follows

3.1 Limitations of Traditional Models

Past studies show that using traditional statistical models and certain algorithms on financial time-series data often fails due to the following reasons:

- **Inflexibility to Non-linear Dynamics:** Traditional models like ARIMA or Naive Bayes usually struggle to handle large datasets or capture the complex relationships in financial markets [1, 2, 3] .
- **Weakness of Distance-Based Algorithms:** Models like K-Nearest Neighbors (KNN) [4] and Support Vector Machines (SVM) [5] are highly ineffective for predicting raw time-series data. Past evaluations showed that KNN had a high Mean Squared Error (MSE) of 61,218.36. Meanwhile, SVM struggled to adjust its hyperplane flexibly, resulting in a massive error of up to 407,690.66 [6].
- **Multicollinearity Issues:** Feeding highly correlated independent variables into a model makes regression coefficients unstable, which significantly reduces the model's accuracy [7].

3.2 Proven Architectures & Solutions

To overcome the limitations mentioned above, successful research often applies more advanced techniques:

- **Advanced Data Preprocessing:** Multicollinearity can be solved using Principal Component Analysis (PCA). This technique reduces the data's dimensions while keeping up to **96.37%** [8] of the original data's variance.
- **Effective Baseline Models:** Linear Regression and Ridge Regression serve as excellent baselines, achieving an MSE as low as 45.39 [6]. Additionally, Lasso Regression is outstanding for feature selection because it can reduce the coefficients of noisy variables to exactly zero [9].
- **The Power of Tree-Based Models:** Random Forest is highly robust against outliers and accurate in identifying trend directions[11]. Advanced algorithms like XGBoost and LightGBM are considered state-of-the-art because they minimize both bias and variance [10]. In one study, XGBoost achieved an Explained Variance Score higher than **0.959** and kept the Mean Absolute Error (MAE) at just 0.0238.
- **Deep Learning Networks:** Capturing long-term relationships requires models like LSTM [12], which achieved a low MSE of 376.603. Furthermore, Hybrid Frameworks (such as combining CNN with Bi-LSTM) improve predictions by blending feature extraction with forward and backward sequential analysis [13].

3.3 Thai Local Contexts

Since the primary objective of this study is to predict the physical gold price in Thailand (as benchmarked by the Gold Traders Association), the model cannot rely exclusively on the global spot gold price (XAU/USD). The pricing mechanism of domestic gold is inherently tied to the local exchange rate and importing costs. Mathematically, the Thai gold price formulation, as officially standardized by the Gold Traders Association and the Thailand Futures Exchange (TFEX) [14], [15], can be expressed as follows:

$$\mathbf{P_{TH}} = \left(\mathbf{P_{XAU}} + \mathbf{C_{prem}}\right) \times \mathbf{E_{FX}} \times \left(\frac{\mathbf{15.244}}{\mathbf{31.1035}}\right)$$

Where:

- $\mathbf{P_{TH}}$: represents the Thai physical gold price per Baht-weight (in THB).
- $\mathbf{P_{XAU}}$: is the global spot gold price (in USD per troy ounce).
- $\mathbf{C_{prem}}$: denotes the premium, including freight, insurance, and local importing costs.
- $\mathbf{E_{FX}}$: is the real-time USD/THB exchange rate.
- $\frac{\mathbf{15.244}}{\mathbf{31.1035}}$: is the standard weight conversion ratio from one troy ounce (31.1035 grams) to one Thai Baht-weight (15.244 grams).

4. Framing the Problem

Gold price volatility makes investment decisions inherently risky, as price movements are influenced by a complex mix of macroeconomic and geopolitical factors that are difficult to predict by intuition alone. To tackle this, the project applies machine learning techniques and frames the problem into two main parts:

1. **Regression** — predicting the closing price of gold on a given day.
2. **Classification** — identifying trading signals to categorize market conditions as buy, sell, or hold.

5. Objectives & Success Criteria

5.1 Project Objectives

- **To develop a gold price prediction model in volatile markets:** To build a Machine Learning model capable of handling gold market volatility. This includes predicting closing prices (Regression) and classifying market trends (Classification).
- **To extract and test alternative features:** To study the effectiveness of using proxy variables, such as the VIX Index and government bond yields, instead of Natural Language Processing (NLP) to capture safe-haven flows during crises.
- **To minimize false signals:** To apply time-series data techniques (Lag Features) and volatility indicators (like ATR) to filter out market noise and prevent fake breakouts.
- **To create an evidence-based investment strategy simulation:** To apply the fully developed and tuned models to backtesting. This will simulate financial returns and assess risks before real-world application.

5.2 Success Criteria

5.2.1 Dimension 1: Academic & Baseline Metrics

- **Baseline Performance:** Baseline classification models, specifically Logistic Regression as the primary competitive baseline and Naive Bayes as the statistical lower bound, must both classify Buy and Sell signals significantly better than random guessing, achieving Accuracy $> 50\%$ and ROC-AUC ≥ 0.55 on the test set. For regression, all linear baseline models must outperform the Persistent Naive Forecast as measured by RMSE and MAE.

- **Non-linear Model Accuracy:** Non-linear models (such as Decision Tree or Random Forest) must show lower error rates (RMSE or MAE) than a Linear Regression model when predicting closing prices.

- **Feature Effectiveness:** Adding proxy variables (VIX, Bond Yield) and volatility features (ATR) must improve the overall performance of the model compared to using only historical gold prices.

5.2.2 Dimension 2: Advanced Machine Learning Metrics

- **Precision (Signal Accuracy):** The Classification model must achieve a Precision score of **55% - 65%** for the "Buy" class. This confirms that the system prioritizes certainty. Missing an opportunity (False Negative) is acceptable and better than generating false buy signals (False Positive) that lead to financial losses.

- **ROC-AUC Score:** The model must be able to clearly distinguish between upward and downward market trends. The Area Under the Curve (ROC-AUC) must be at least **0.65 - 0.70**. This proves the model has real predictive power and is not just memorizing the training data (Overfitting).

5.2.3 Dimension 3: Financial & Risk Management Metrics

- **Net Profit:** After backtesting and deducting trading costs and fees, the simulated investment portfolio must show a positive total return.

- **Profit Factor:** The ratio of Gross Profit to Gross Loss within the system must be greater than **1.5**.
- **Maximum Drawdown (MDD):** The system must limit risks during severe market volatility. The maximum accumulated loss from a peak (MDD) must remain at an acceptable level (e.g., no more than **15-20%** of the portfolio). This proves the model is safe enough for practical use.

6. Formulating Hypotheses

6.1 Hypothesis 1 : Baseline Model Performance

This project establishes minimum performance bounds separately for both tasks before advancing to complex models.

For the Classification task, Logistic Regression serves as the main competitive baseline, as it imposes fewer distributional assumptions and is well-suited to linearly separable financial signals. Naive Bayes acts as the statistical lower bound, since its independence assumption fundamentally contradicts the high inter-correlation observed among price-derived features such as XAU/USD log returns and VIX Z-Score. Both models must classify Buy and Sell signals significantly better than random guessing, achieving Accuracy $> 50\%$ and ROC-AUC ≥ 0.55 on the test set (2023–2025). This confirms that the selected feature set carries sufficient predictive power before progressing to advanced architectures.

For the Regression task, Linear Regression serves as the main baseline, establishing the performance ceiling of purely linear assumptions. Ridge and Lasso Regression serve as supplementary regularized baselines. Ridge is expected to outperform plain Linear Regression by reducing coefficient instability caused by multicollinearity among features, while Lasso is expected to perform implicit feature selection by shrinking the coefficients of low-signal variables toward zero. All three models must outperform the Persistent Naive Forecast (predicting tomorrow's price equals today's price), which serves as the dummy lower bound, as measured by RMSE and MAE on the test set (2023–2025).

6.2 Hypothesis 2 : Time-Series and Volatility Feature Management

Creating and incorporating lag features (e.g., historical price returns from the past 7–15 trading days) alongside volatility indicators such as the Average True Range (ATR) and 30-Day Rolling Realized Volatility will improve the model's ability to distinguish genuine trend reversals from temporary price fluctuations. It is hypothesized that models trained with these time-series features will yield a measurably lower False Positive rate on Buy signals compared to models trained on contemporaneous features alone, thereby reducing the risk of acting on short-lived noise in volatile market conditions.

6.3 Hypothesis 3 : Macroeconomic Factors and Crisis Proxies

During periods of elevated geopolitical tension and financial uncertainty, market participants tend to reallocate capital toward safe-haven assets such as gold. It is hypothesized that incorporating the VIX Index (as a proxy for US equity market fear) and US Government Bond Yields (as a proxy for risk-free return opportunity cost) as exogenous variables will capture these safe-haven capital flows effectively. This approach is expected to improve model performance during crisis periods without requiring a Natural Language Processing (NLP) pipeline to directly interpret news sentiment. Models trained with these macroeconomic proxies are expected to achieve higher ROC-

AUC scores on volatile sub-periods of the test set compared to models using only price-derived features.

6.4 Hypothesis 4 : Algorithm Complexity

Because the relationship between gold prices and macroeconomic variables is inherently non-linear and subject to regime changes, it is hypothesized that model performance will improve progressively as algorithmic complexity increases. Specifically, Ridge and Lasso Regression are expected to outperform plain Linear Regression by addressing multicollinearity and irrelevant features. Subsequently, non-linear tree-based models — namely Decision Tree Regression and Random Forest Regression — are expected to outperform all linear variants by capturing interaction effects and threshold-based dynamics that linear models cannot represent. This improvement will be evaluated using RMSE and MAE on the test set (2023–2025), with the expectation that Random Forest will achieve an RMSE at least 10% lower than the Linear Regression baseline under identical feature conditions.

7. Data Assessment

7.1 Feature Categories & Economic Rationale

Based on financial market dynamics, candidate features were organized into five structural categories. Each category represents a distinct economic mechanism through which external forces transmit their effects onto gold prices.

Feature Category	Variables & Indicators	Economic Rationale
Price Technical Indicators	Moving Averages (MA 7, 15, 30), RSI, MACD, Bollinger Bands, ATR, Rate of Change (ROC), Garman Klass Volume	Used to filter short-term market noise, identify momentum strength, detect overbought/oversold conditions, and measure price range volatility
Currency & Rates	EUR/USD, USD/THB, US Dollar Index (USDI), US Bond Rates, XAU/USD	Since gold is priced in USD, a stronger dollar typically suppresses gold prices inversely. Bond yields reflect the opportunity cost of holding a non-interest-bearing asset such as gold
Commodity Complex	United States Oil Fund (USO), Brent Crude Oil, Silver Futures (SF), iShares Silver Trust (SLV), Platinum, Palladium, Rhodium	Oil prices are deeply linked to inflation expectations, which directly affect demand for gold as an inflation hedge. Silver exhibits a high statistical co-movement with gold
Gold Miners Dynamics	Gold Miners ETF (GDX), Eldorado Gold Corporation (EGO)	Mining equities reflect the industry's forward-looking view on gold prices, as well as supply-side pressures from production costs and extraction volumes
Equity Market Linkages	S&P 500 (SPX), Dow Jones Index (DJ), NASDAQ-100 (NDX), VIX, GVZ	Serves as a proxy for risk-asset sentiment. Sharp equity market declines historically trigger capital rotation into gold as a safe-haven asset

7.2 Data Scope & Sourcing

This section defines the scope and origin of all raw data considered for this project prior to any screening or processing.

- **Timeframe:** January 2, 1990 – December 31, 2025
- **Target Variable:** Daily closing price of 96.5% purity gold bullion, sourced from the Gold Traders Association (GTA) at https://www.goldtraders.or.th
- **Primary Benchmark:** Global Gold Spot price (XAU/USD), sourced from MetaTrader 5
- **Currency Data:** USD/THB exchange rate, sourced from MetaTrader 5
- **Commodity Data:** Brent Crude Oil, WTI, Silver, and related instruments sourced from http://investing.com
- **Volatility & Sentiment Indices:** VIX and GVZ sourced from https://www.cboe.com/tradable-products/vix/vix-historical-data
- **Equity Indices:** S&P 500, NASDAQ-100, and Dow Jones Index sourced from public financial data providers

- **Gold Miners Data:** GDX and EGO sourced from public equity market feeds
- **Supplementary Datasets:** Where applicable, the Hugging Face Datasets library will be used to access publicly available financial or economic datasets that may support model benchmarking

7.3 Data Quality & Pre-processing

As this project is currently in the Discovery Phase, the full preprocessing pipeline will be designed and documented in Phase 2. However, the following key preprocessing principles have been identified based on the nature of financial time-series data.

7.3.1 Stationarity & Transformation

Raw price series are inherently non-stationary, as confirmed by preliminary Augmented Dickey-Fuller (ADF) testing. All price-based variables will therefore be transformed into stationary representations — primarily via Log Return transformation — prior to model training. Volatility features will be expressed as rolling statistical measures to preserve their temporal structure.

7.3.2 Data Alignment

The five feature categories originate from markets with differing trading hours and holiday calendars. An Inner Join strategy based on the shared trading date index will be applied to ensure that only dates with complete observations across all sources are retained, preventing the model from learning on incomplete cross-market information.

7.3.3 Train/Test Split Strategy

Because this project uses time-series data, random splitting will be strictly avoided to prevent data leakage. The primary strategy will be a Time-Based Split, where data from 1990–2022 serves as the training set and data from 2023–2025 serves as the test set, simulating a real-world forecasting scenario. Walk-Forward Validation will additionally be considered during hyperparameter tuning to evaluate model stability across distinct market regimes, including the COVID-19 period and subsequent geopolitical volatility cycles.

7.4 Initial Correlation Assessment

A preliminary correlation screening was conducted to evaluate which candidate variables demonstrate a statistically meaningful relationship with Thai gold bullion prices. The key findings are as follows.

Strong Positive Correlation: Thai gold bullion prices show a strong positive correlation with XAU/USD returns and USD/THB log returns. This is structurally consistent with the Thai gold pricing mechanism described in Section 3.3, where both the global benchmark and the exchange rate directly determine domestic prices.

Volatility Indicators: VIX and Realized Volatility do not exhibit a clear linear relationship with gold prices under normal market conditions. However, Kernel Density Estimation plots reveal that during high-volatility regimes, the distribution of gold returns widens significantly — confirming that these variables carry regime-dependent information that non-linear models may be able to exploit.

Low Correlation Factors: Brent Crude Oil and WTI demonstrated low daily correlation with gold prices. The transmission mechanism between energy prices and safe-haven demand operates through inflation indices, which introduces a lagged effect incompatible with daily price prediction. These variables were therefore excluded from the preliminary feature set.

Preliminary Feature Selection: Based on this initial screening, XAU/USD Log Returns, USD/THB Log Returns, VIX 30-Day Z-Score, and 30-Day Rolling Realized Volatility were identified as the primary candidate inputs for model development. These variables collectively cover the price signal, currency translation, and market fear dimensions. Final feature selection and importance ranking will be conducted formally during Phase 2 using correlation matrices, Variance Inflation Factor (VIF) analysis, and tree-based feature importance scores.

8. Resource Assessment & Roles

To execute the project effectively, we have assessed our resource readiness and defined an initial team structure to ensure systematic and efficient management. The details are as follows:

8.1 Technological Resources

The infrastructure and tools for the initial phase focus on agility and team collaboration. These include:

- **Environment & Compute:** We use Google Colab as our primary coding environment because it supports real-time collaboration and reduces personal hardware limitations.
- **Version Control:** We use GitHub to store code, control development versions, and allow members to integrate their work systematically.
- **Project Management:** We use GitHub's Kanban board to track task progress. This makes it easy to manage sub-tasks in each phase and effectively monitor the team's workflow.

8.2 Data Resources

The project uses data covering the period from 1990 to the end of 2025 (January 1, 2010, to December 31, 2025). The data sources are as follows:

- **Domestic Gold Bullion Prices:** Sourced from the Gold Traders Association (GTA) website (https://www.goldtraders.or.th).
- **Global Gold Spot (XAU/USD):** Sourced from MetaTrader 5.
- **Currency Exchange Rate (USD/THB):** Sourced from MetaTrader 5.
- **Brent Crude Oil and WTI:** Sourced from http://investing.com.
- **VIX and GVZ:** Sourced from https://www.cboe.com/tradable-products/vix/vix-historical-data.
- **Supplementary Datasets:** Where applicable, the Hugging Face Datasets library is used to access and integrate publicly available financial or economic datasets that may support model training and benchmarking.

8.3 Time Resources

- **Expected Delivery Date:** May 12, 2026

8.4 Human Resources & Roles

To ensure clear operations in Phase 1, core responsibilities have been assigned to team members. *(Note: A single member can hold multiple roles, and some roles can be shared among members depending on the workflow.)*

The role assignments are as follows:

- **Project Manager:** Athiphat Sunsit
    - **Main Duties:** Acts as the center for planning and managing the team's daily tasks. Organizes and tracks progress via the Kanban Board, ensures everyone meets milestones, and coordinates or solves immediate problems to keep the project moving forward.
- **Business Intelligence:** Napattheera Laokoamhung
    - **Main Duties:** Understands project goals, analyzes success metrics, and determines the presentation format that best meets the practical needs of the project.
- **Data Engineer:** Theepob Rattanapsapsiri
    - **Main Duties:** Plans the data pipeline structure, evaluates data sources, and prepares data extraction into the analysis system.
- **Database Administrator:** Patcharapon Malaisri
    - **Main Duties:** Evaluates and designs the data storage architecture to ensure it can fully support the datasets used in the project.
- **Data Scientists:** Purich Ampawa and Panithan Thantue
    - **Main Duties:** Work together to frame problems, formulate hypotheses, assess data relevance, and plan the analytical techniques or models for the next phase.

Beyond their main responsibilities, every team member supports all aspects of the project. The primary goal is for everyone to learn. This approach provides an opportunity for all members to rotate and experience every position, ensuring that everyone fully understands how to develop a Data Science project using Machine Learning according to the project's objectives.

9. Project Risks

In developing the gold price forecasting model for the Gold ML Prediction project, we have evaluated three main risks that could impact the model's performance:

9.1 Risk from Not Using Natural Language Processing (NLP) for News Analysis

- **Cause:** The project uses the VIX index and government bond yields as proxies to capture money flows, rather than using NLP techniques to directly analyze the context of news articles.
- **Impact:** The model may respond slowly to sudden emergencies (such as sudden declarations of war or new political policies). It must wait for market panic to reflect in the VIX or GVZ indices first, which could create a disadvantage in real-time decision-making.
- **Mitigation:** The team will focus on using models that effectively capture volatility and evaluate the impacts reflected in lag features. This ensures the model can respond to changing statistical trends as quickly as possible.

9.2 Risk from Unprecedented Historical Events (Black Swan Events)

- **Cause:** The model is trained using historical data from 2010 to the end of 2025.
- **Impact:** If a new type of economic crisis or unprecedented geopolitical instability occurs in 2026, the historical relationships between the analyzed variables (such as XAU/USD and volatility) may change. This could lead to a decrease in the model's accuracy.
- **Mitigation:** The system is designed to continuously perform backtesting and closely monitor the Maximum Drawdown (MDD). If the simulated portfolio experiences losses exceeding an acceptable threshold, it will serve as a clear signal for the team to perform Model Retraining.

9.3 Risk from Differences in Market Trading Hours

- **Cause:** Global gold prices and reference indices are traded almost 24 hours a day, whereas domestic gold bullion prices follow standard Thai market opening and closing hours.
- **Impact:** Even though we use an "Inner Join" technique to resolve mismatched market holidays, price volatility that occurs outside of Thai trading hours (such as overnight) may not be perfectly reflected in the Thai closing price.
- **Mitigation:** Converting the raw price data into Daily Logarithmic Returns will help adjust for data non-stationarity and, to some extent, reduce the impact of sudden price jumps when the market opens.

10. Conclusion

This report documents the foundational work completed during Phase 1 of the Gold Price Prediction project. The primary focus of this phase was not yet to build models, but to establish a clear and well-reasoned direction before any technical implementation begins.

The team began by framing the problem into two distinct tasks — regression for predicting daily gold closing prices, and classification for identifying Buy and Sell trading signals. From this foundation, four research hypotheses were formulated to guide model selection and evaluation across both tasks, covering baseline performance bounds, the role of time-series features, the use of macroeconomic proxies, and the expected progression in accuracy as model complexity increases.

For data, the team expanded the candidate feature set beyond simple price history into five structured categories: Price Technical Indicators, Currency and Rates, Commodity Complex, Gold Miners Dynamics, and Equity Market Linkages. A preliminary correlation screening confirmed that XAU/USD Log Returns, USD/THB Log Returns, and VIX are the most structurally and statistically relevant variables for predicting Thai gold prices at this stage. Final feature selection will be formalized in Phase 2.

Three key project risks were also identified — the absence of NLP-based news analysis, exposure to Black Swan events, and misalignment across market trading hours — each with a defined mitigation strategy.

Overall, Phase 1 has established a clear problem definition, a structured data strategy, and a testable set of hypotheses. These outputs provide a solid foundation for the modeling and evaluation work that will follow in the next phase.

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